



Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

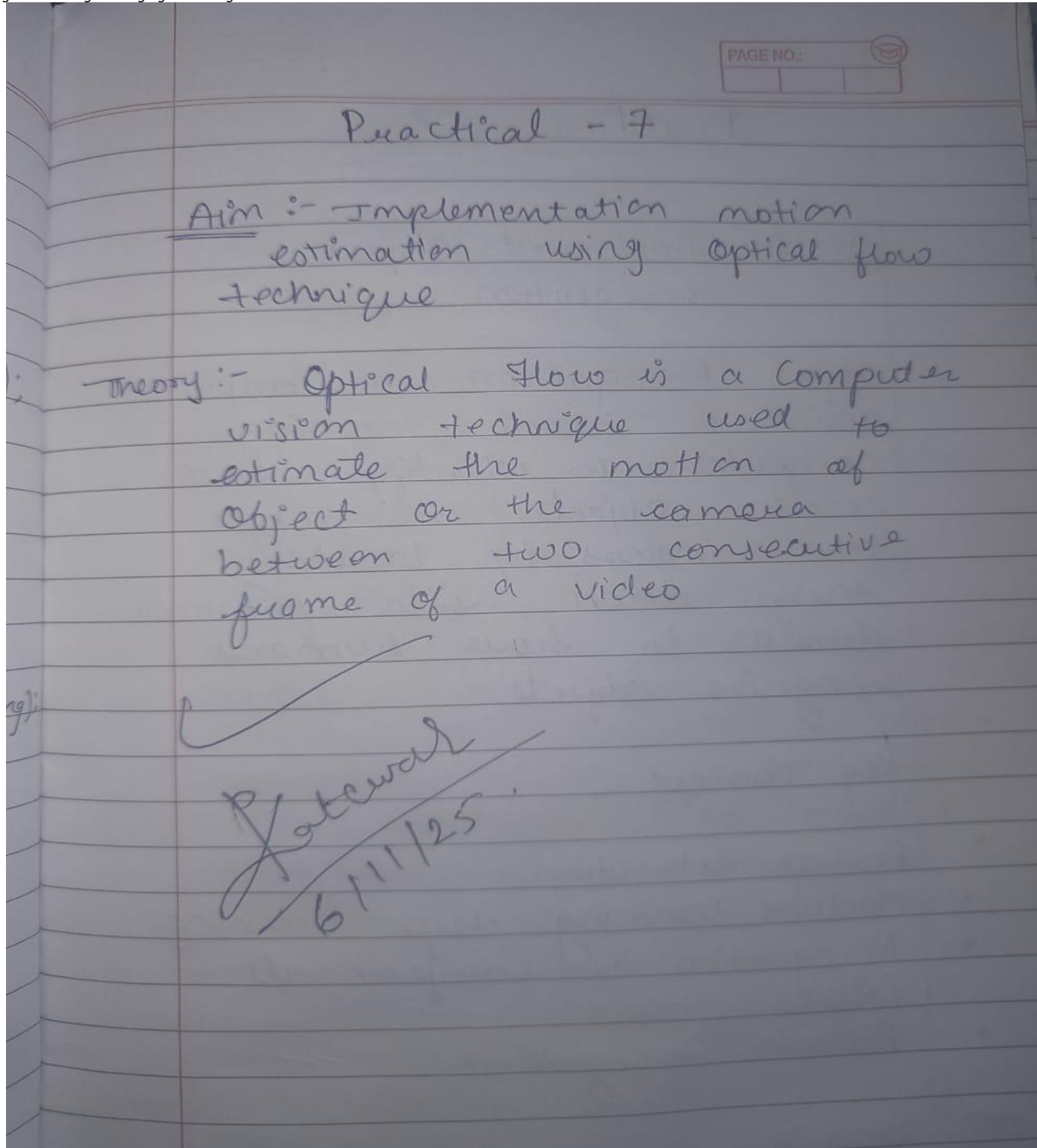
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Ganesh Pandile

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2024-25 (ODD)	Course Name	Computer vision Lab
Semester	5	Course Code	CT
Roll No	46	Name of Student	Ganesh Pandile

Practical Number	7
Course Outcome	Upon successful completion of the course the students will be able to 1. Apply image enhancement and smoothing techniques to improve image quality for further analysis. 2. Extract meaningful features from images using descriptors such as HOG and SIFT. 3. Implement and evaluate modern object detection methods including YOLO and R-CNN. 4. Analyze and develop solutions for motion estimation, object recognition, and facial expression recognition using classical and learning-based methods.
Aim	Implement motion estimation using optical flow technique.
Problem Definition	Given two or more consecutive image frames from a video sequence, the goal is to estimate the motion vectors (optical flow) that describe the movement of each pixel or region between frames. These motion vectors should accurately capture both the magnitude and direction of motion.
Theory (100 words)	Motion estimation is a fundamental problem in computer vision that involves determining the movement of objects or features between consecutive frames in a video sequence. Accurate motion estimation is essential for numerous applications such as video compression, object tracking, autonomous navigation, gesture recognition, and 3D scene reconstruction. One of the most widely used approaches for motion estimation is the optical flow technique. Optical flow represents the apparent motion of brightness patterns in an image sequence, providing a dense field of motion vectors that describe the displacement of pixels over time.
Procedure	Input: Two consecutive video frames F_t and F_{t+1}. Convert both frames to grayscale.



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and
Execution

(100 Words)

Compute optical flow between frames using an algorithm (e.g., Farnebäck or Lucas–Kanade).

Obtain motion vectors (magnitude and direction) for each pixel or feature point.

Visualize motion using arrows or color-coded flow map.

Output: Estimated motion field representing object movement between frames.

Code:

Code:

```
# Install dependencies (OpenCV)
!pip install opencv-python-headless
```

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import files
```

```
# Step 1: Upload video file
print("Upload a video file for motion estimation (e.g., mp4):")
uploaded = files.upload()
```

```
# Get the uploaded file name video_file
video_file = list(uploaded.keys())[0]
```

```
# Step 2: Optical flow parameters and initialization
feature_params = dict(maxCorners=100,
                      qualityLevel=0.3,
                      minDistance=7,
                      blockSize=7)
```



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```
lk_params = dict(winSize=(15, 15),
maxLevel=2,
criteria=(cv2.TERM_CRITERIA_EPS |
cv2.TERM_CRITERIA_COUNT, 10, 0.03))

# Colors for visualization color = np.random.randint(0, 255, (100,
3))
# Step 3: Read video and process frames cap
= cv2.VideoCapture(video_file)

ret, old_frame = cap.read() if
not ret:
    print("Failed to read video")
cap.release()    exit()

old_gray = cv2.cvtColor(old_frame, cv2.COLOR_BGR2GRAY)

# Detect initial points to track p0 =
cv2.goodFeaturesToTrack(old_gray, mask=None,
**feature_params)

# Create a mask image for drawing optical flow tracks mask
= np.zeros_like(old_frame)

frame_count = 0 while
True:
    ret, frame =
cap.read()    if not ret:
break

    frame_gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Calculate optical flow    p1, st, err =
cv2.calcOpticalFlowPyrLK(old_gray, frame_gray, p0, None,
**lk_params)
```



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```
# Select good points
if p1 is not None:
    good_new = p1[st == 1]
    good_old = p0[st == 1]

    # Draw the tracks      for i, (new, old) in
enumerate(zip(good_new, good_old)):
    a, b = new.ravel()      c, d = old.ravel()      mask =
cv2.line(mask, (int(a), int(b)), (int(c), int(d)), color[i].tolist(), 2)
frame = cv2.circle(frame, (int(a), int(b)), 5, color[i].tolist(), -1) img
= cv2.add(frame, mask)
    # Show result in colab
    print(f'Frame {frame_count}')
cv2_imshow(img)

# Update previous frame and points
old_gray = frame_gray.copy()  p0 =
good_new.reshape(-1, 1, 2)  else:
    break

frame_count += 1

cap.release() print("Done processing video.")
```

Output:

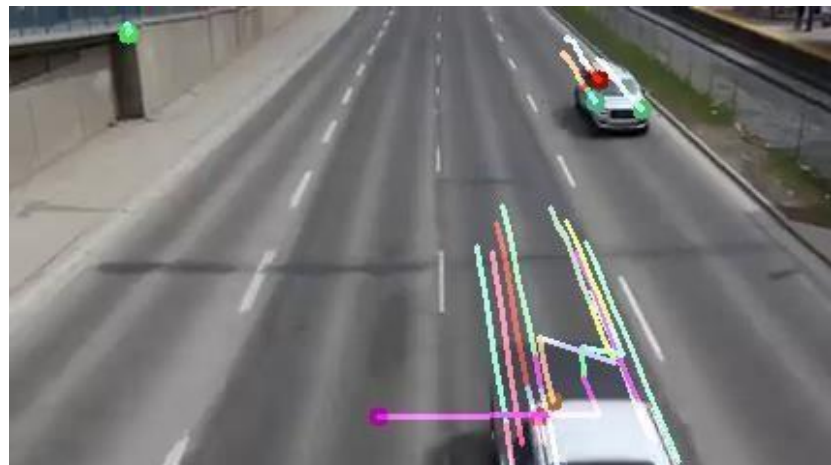
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Output Analysis	1. Visual Output • The system generates a motion vector field or a color-coded flow map showing how each pixel (or feature point) moves • Direction of motion is represented by color hue, and magnitude (speed) is represented by brightness or vector length. • When tested on sample videos: ○ Static regions show little or no color change (no motion). ○ Moving objects appear in distinct colors indicating their movement direction and speed 2. Quantitative Analysis (optional if you have data) • The magnitude of optical flow vectors corresponds to object velocity. • Accuracy can be evaluated by comparing estimated motion with ground truth or by visual inspection of consistency. • Performance depends on: ▪ Frame rate (higher is smoother) ▪ Lighting changes (may affect accuracy)

	• Motion speed (very fast motion may cause blur or flow errors) 3. Observations • The Farnebäck method provides dense and smooth flow, suitable for full-frame motion visualization. • The Lucas–Kanade method gives sparse but accurate feature tracking, ideal for object tracking applications. • The optical flow technique effectively captures motion direction and magnitude but may struggle with:
Link of student Github profile where lab assignment has	https://github.com/ganeshpandile/Computer-Vision.git



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Conclusion	The implemented optical flow algorithm successfully estimates and visualizes motion between consecutive video frames. The output demonstrates that optical flow can effectively represent both direction and speed of moving objects, making it useful for tasks like object tracking , video analysis , and autonomous navigation ..
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